Math 9 Fall 2022 — Lecture #5 Worksheet
Corresponding Outcomes: EO3, EO4, GOMa1, GOMa2

This is an in-class worksheet. You do not need to turn in anything.

Try to understand what the following provided functions are doing just by reading the code,i.e., without evaluating them in MATLAB.

1. Consider the following function:

```
function [c] = countA(v)
c = 0;
for i = 1:length(v)
    if v(i) < 4
        c = c+1;
end
end
```

   Write a function countB which has the same behavior using a single line of code (3 total lines if you count the "function" line and the "end" line). Your function should not have any loops or if statements in it. You should assume v is a row vector. (Hint. Use a logical vector.)

```
function [c] = countB(v)
    ???
end
```

2. Consider the following function:

```
function [b] = two_equalA(v)
for i = 1:(length(v)-1)
    if v(i) == v(i + 1)
        b = true;
    else
        b = false;
    end
end
end
```

   Write a function `twoequalB` which has the same behavior using a single line of code. You should assume the input v is a row vector. (Warning. The original function might not work as you would expect from the name.)

```
function [b] = two_equalB(v)
???
end
```

3. Consider the following function:

```
function [w] = replaceA(v,a)
w = [];
```

```
for x = v
    if x > a
        w = [w,x];
    else
        w = [w,0];
    end
end
end
```

Write a function `replaceB` which has the same behavior using a single line of code. You should assume v is a row vector and a is a number.

```
function [w] = replaceB(v,a)
???
end
```